

Christopher McLoughlin

463-201-6767 | cmcloug@purdue.edu | github.com/cmcloug | linkedin.com/in/chrismclou

Education

Purdue University, College of Science

Bachelor of Science, Computer Science with Minor in Communications

Anticipated May 2028

West Lafayette, IN

Experience

Interactive Brokers LLC

Software Development Intern, Enterprise Architecture

June 2026 – August 2026

Greenwich, CT

- Built an internal AWS FinOps tool in Python, querying cost and usage data with AWS Athena and SQL across all organizational accounts and generating formatted Excel reports for any user-selected date range, cutting reporting time from 4–6 hours to 4 minutes and eliminating manual errors.
- Engineered a reusable AWS savings analysis tool in Python, using Athena and SQL to compare usage and cost trends across thousands of instances and numerous accounts against Reserved Instance and Savings Plan opportunities, identifying significant reductions in monthly on-demand spend.
- Led a team in automating cleanup of unused AWS sandbox resources with Python, owning project timelines, meetings, task tracking, and stakeholder communication across multiple teams.
- Supported IBKR's procurement process by conducting vendor outreach, coordinating product demos, and building comparison sheets that informed internal platform selection.

Purdue University & Feenix Group

Undergraduate Data Science Researcher

August 2025 – May 2026

Lafayette, IN

- Partnered with Feenix Group and a team of 8 developers to design and deploy a real-time analytics dashboard tracking engagement and monetization metrics across Roblox games.
- Engineered a relational database and scheduling system that automates periodic data collection from public APIs, enabling continuous time-series tracking with no manual intervention.
- Implemented a timestamp retrieval function to pull historical snapshots for on-demand analytics and period-over-period comparisons.
- Used SQL and R for data aggregation and visualization, uncovering player retention and performance patterns that informed product strategy.
- Improved scraping efficiency by 44% by optimizing data requests and query scheduling.

Project

interview.you

- Developed a full-stack AI mock interview platform with a React and Tailwind frontend and a FastAPI backend, featuring a voice-based AI interviewer that listens and responds in real time; amassed 500+ user accounts.
- Personally designed the database schema in PostgreSQL with SQLAlchemy, modeling users, interview sessions, code submissions, and progress history, and used Redis for caching and session state.
- Built a sandboxed environment that executes and grades real code against hidden test cases, plus camera-based analysis of posture and eye contact.
- Added resume and cover letter review and long-term progress tracking with personalized recommendations.

Leadership Experience

Purdue CS Undergraduate Student Board

President

- Manage a 25-member board and oversee 13 concurrent initiatives, including mentorship, tutoring, and technical workshops, that draw hundreds of student participants each year; maintain a close working relationship with faculty and the department.

Eli Lilly Summer Science Camp

Site Supervisor

- Managed 42 counselors and 250+ campers, ensuring safety and smooth daily operations; served as the main communication link between parents and counselors.